The Price of Intelligence

Hyper 5 spent $1.1 trillion on capex in the past five years, with $5.3 trillion more expected by 2030

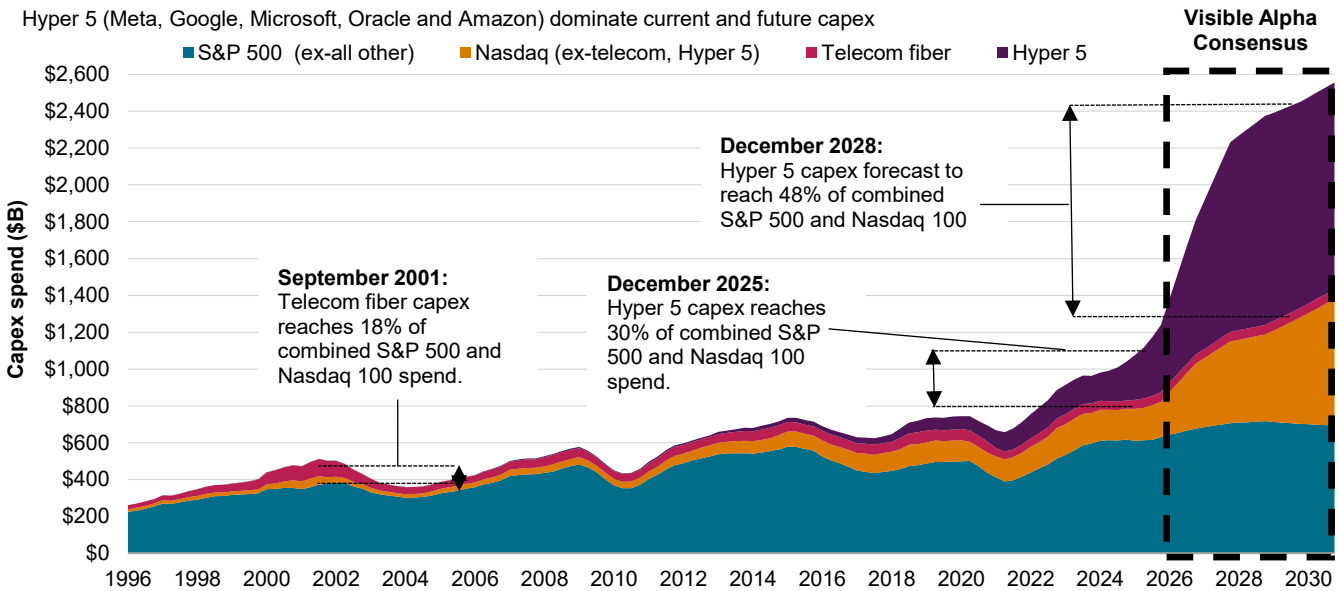
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The artificial intelligence (AI) build-out has reached historically extreme investment intensity and unprecedented concentration, but it has not yet developed the financing characteristics that made the dot-com build-out fragile. Measured by capital expenditure (capex) relative to depreciation (dep), investment intensity has exceeded prior cycle peaks, including those of the dot-com era and Great Recession (GR). This cycle is being driven disproportionately by hyperscalers, which have collectively spent $1.1 trillion in capex over the past five years. Visible Alpha sell-side consensus adds $5.3 trillion more by 2030. The key risk is the transition from cash- to debt-funded investment before returns validate the spending.

Figure 1

Eight Years of AI-era capex to exceed the prior 26 years combined



As of Aug. 24, 2026.
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- **Lessons learned.** Capex cycles typically move through four phases: build, stretch, overshoot and digest. Investment intensity rises when expected earnings translate to higher returns on new capital (build). This has historically persisted despite tighter monetary policy (stretch). The cycle turns when the next investment return no longer exceeds the cost of financing (overshoot). Markets then undergo a period of lower investment (digest).
- **Cash wedge.** S&P 500 cash-to-capex reached an unprecedented 330% (3.3x) in 2021, three times the level at the dot-com peak, and has been unwinding since, partly to fund the build-out.
- **Concentration**. Visible Alpha sell-side consensus is for the Hyper 5 (Amazon, Alphabet, Microsoft, Meta, Oracle) to reach 52% of total S&P 500 capex by 2028, versus 13% and 30% in 2020 and 2025, respectively.
- **Valuations**. From an earnings multiple perspective, the rest of the S&P looks historically more expensive than the hyperscaler companies spending the capex as the market awaits increased earnings from AI efficiency gains.

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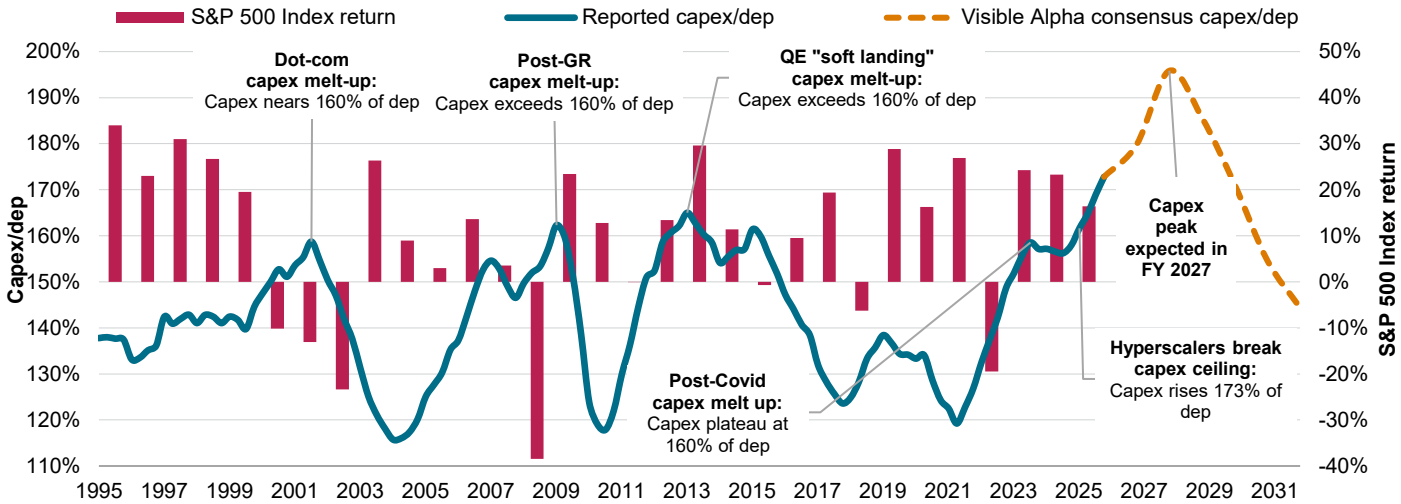
1. Anatomy of a capex melt-up

The capex-to-dep ratio provides a simple measure of reinvestment intensity (Figure 2, blue line). Depreciation approximates the ongoing consumption of the existing asset base, so a company investing only enough to maintain current operations would have capex roughly equal to depreciation, or a ratio near 100%. Readings materially above 100% indicate that firms are collectively expanding the capital base faster than it is being consumed.

Figure 2

S&P 500 capex melt-up surpasses dot-com bubble and Great Recession levels

Bottoms-up capex to dep of the asset base and total index return, S&P 500



As of Aug. 17, 2026.
QE = quantitative easing.
Source: S&P Global Market Intelligence.
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Viewed over six decades, these investment cycles follow a recurring pattern that we characterize in four phases: **build, stretch, overshoot and digest**.

Build. The build phase begins around the troughs in reinvestment intensity (blue line in Figure 3). These troughs mark a turning point: After a period of weak or declining investment, firms collectively begin to conclude that the expected return on new capacity once again warrants increased capital spending. Monetary policy is typically accommodative, keeping the cost of capital low and making a wider range of projects economically viable. Capex begins to turn higher.

Stretch. Build becomes stretch when the pace of investment accelerates despite a higher cost of capital. In Figure 3, these are the rising segments of the blue capex-to-dep line. Importantly, they frequently overlap with the red-shaded periods of rising fed funds rates. The historical pattern suggests that the Federal Reserve responds to the acceleration of investment intensity and increases the risk-free component in the cost of capital.

Importantly, tighter policy does not immediately stop the build-out. Capex continues to rise amid higher policy rates. The reason is economic: Firms will continue making incremental investments as long as marginal returns exceed marginal costs. Fed tightening raises the hurdle rate and stretches the economics of the cycle, but it does not by itself cause investment to turn.

Overshoot. The cycle reaches overshoot when earnings can no longer support the increasingly aggressive expectations embedded in the investment boom. With the exception of the QE-era[1] cycle, each major capex peak in Figure 3 occurs in, or adjecent to, a year marked by a year-over-year earnings contraction, shown by the purple circles.

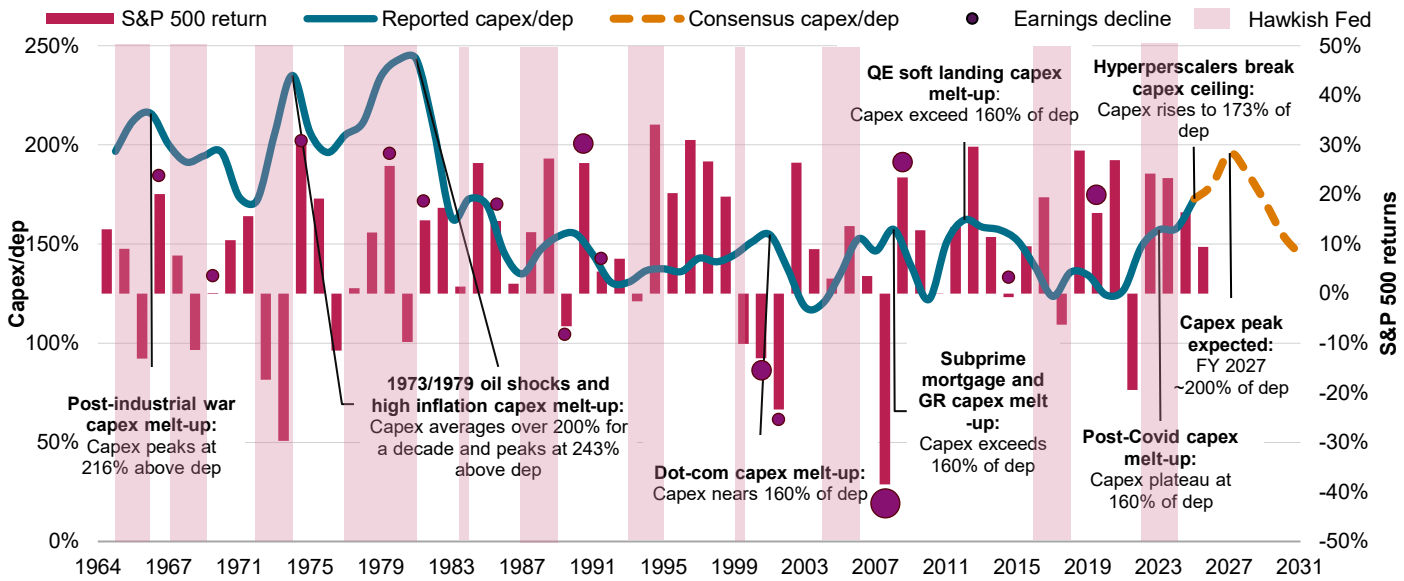
This is the point at which the investment equation begins to fail. On one side, tighter monetary policy has already raised the cost of capital. On the other, deteriorating earnings reduce the expected return on incremental investment. What had continued to pencil during Stretch no longer does. The cycle has extended beyond what underlying cash flows can validate, and new spending begins to lose its economic justification. Financial markets can bring forward the same adjustment. A sharp repricing in equities or credit raises the market-implied cost of capital.

Digest. Once the economics of incremental investment break, the economy enters digest. New capex slows while the capital accumulated during the prior build and stretch phases remains in place. Depreciation continues, weaker projects may be impaired, and companies must allow demand and cash flow to catch up with the capacity already built.

Figure 3

S&P 500 capex melt-up reaches levels not seen since the early 1980s

Bottoms-up capex to depreciation of the asset base, S&P 500



As of Aug. 17, 2026.
Hawkish Fed = +0.5% yoy target fed funds increase.
Source: S&P Global Market Intelligence.
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Monetary policy typically becomes less restrictive during this phase. The Federal Reserve can "take its foot off the brake" as investment cools, earnings weaken and inflationary pressure associated with the build-out subsides. Lower rates reduce the hurdle rate and eventually help create the conditions for the next build phase.

The relationship is therefore not simply that higher interest rates cause lower capex. The historical sequence is more nuanced. Firms build when expected returns justify greater investment. They stretch as capex accelerates even while

[1] The 2013 capex peak coincides with the first taper of QE in December 2013 and was a form of monetary tightening that did not reflect in a higher fed funds target rate. https://www.federalreserve.gov/monetarypolicy.htm

the cost of capital increases. They overshoot when earnings can no longer keep pace with the expectations embedded in that spending. And they digest once the expected return on the next dollar of investment falls below its cost of capital.

The late 1990s and early 2000s dot-com technology boom illustrates why the sequencing matters. Equity markets capitulated before capex did. As market performance deteriorated, monetary policy shifted toward easing, removing part of the financing constraint even as investment continued without sufficient earnings support. Capex therefore remained elevated after the equity signal had already turned, while telecom capacity continued to be built against demand that had not materialized. The eventual adjustment was severe: investment peaked well after equities, earnings deteriorated, and significant market value in global telecom was lost between 2000 and 2002.

Figure 3 therefore suggests that the key signal is not the level of capex alone. It is the interaction between the rate of investment growth, the cost of capital and the earnings required to justify that investment. That framework provides the historical lens through which to evaluate the current AI-driven build-out.

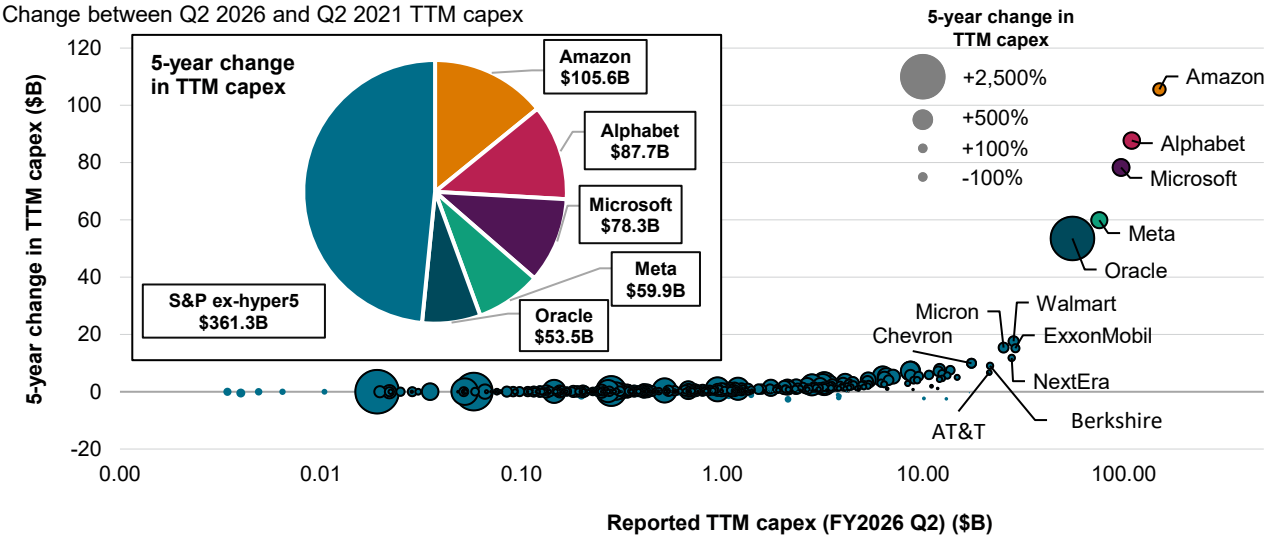
2. The AI build-out

The Hyper 5 are forecast to spend $5.3 trillion in capex between 2026-2030 (versus $1.1 trillion from 2020-2025) and already account for 51.6% of the S&P 500's entire increase in annual capex over the past five years (Figure 1). As a share of the combined S&P 500 & Nasdaq 100[2], Hyper 5 capex went from single-digit in 2019 to 30% by the end of 2025, and is on pace for 48% by 2028 on current Visible Alpha consensus, a concentration the telecom build-out never approached as fiber capex peaked at 19% (Figure 4).

Figure 4

Hyperscalers drive capital investment to 30-year high

Change between Q2 2026 and Q2 2021 TTM capex



As of Aug. 24, 2026.
TTM = trailing 12 months.
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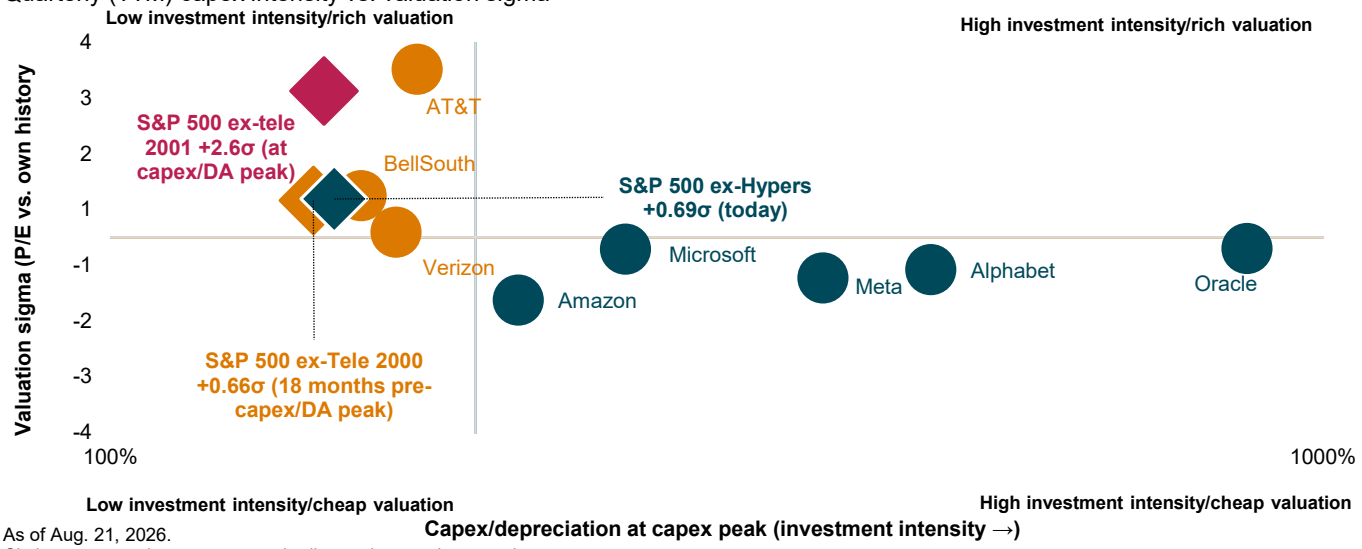
[2] The Nasdaq 100 was added to the S&P 500 to make the analysis universe more comparable to the dot-com era.

But how expensive is the rest of the market right now (Figure 5), and how does that compare with the dot-com era at the same stage of its boom? We measure the S&P 500 excluding telecoms and excluding Hyper 5 at the same moment in their spending cycle, not at a single date. The S&P 500 ex-telecoms did eventually look wildly overpriced (+2.6, meaning their price-to-earnings ratio sat 2.6x standard deviations above its own historical norm, a genuine extreme). But that reading only appeared after the spending had peaked in 2001 and profits had already fallen about 60%. When profits collapse, the price-to-earnings ratio shoots up automatically, even as the share price falls. So that late reading made them look far more expensive than they actually were on the way up.

Figure 5

Hyper 5 sit where the dot-com cohort sat before its capex/depreciation peak

Quarterly (TTM) capex intensity vs. valuation sigma



As of Aug. 21, 2026.
Circles = companies at capex peak; diamonds = market ex-cohort.
Source: S&P Global Market Intelligence .
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The fair comparison is to look at the market at the same point the Hyper 5 are at today, about 18 months before spending was expected to peak, with profits still healthy. At that equivalent moment, the S&P 500 ex-telecoms was only modestly above its own norm (+0.6 sigma, a little over half a standard deviation above its normal price to earnings [P/E]); expensive but hardly a bubble. The S&P 500 ex-Hyper 5 today sits in almost exactly the same place (+0.7 sigma), on a like-for-like basis, today's market is not pricing the AI boom as exuberantly as it was pricing the internet boom on the way into 2000. Looking individually at the Hyper 5, they are spending enormous sums, but investors are not paying a wildly stretched price for them either, with current sigmas all cheaper than historical norms (-0.2 to -1.1 sigma) versus the more expensive telecoms[3] above their historical norms (with AT&T the standout at +3 sigma). If anything, the dot-com era teaches us the danger was never the price, but what happened to profits afterward.

The infrastructure build-outs from both eras also rhyme. The Hyper 5 are laying capacity at a pace that echoes the telecom carriers of the late 1990s, when WorldCom, Global Crossing, Qwest and others raced to build a fiber-optic network for demand still mostly on paper. Nvidia occupies a similar hardware-supplier role to the one Cisco played

[3] Only three telecoms had a usable P/E at their capex peak. The pure-plays (Level 3, Broadwing, Global Crossing) had no earnings; Sprint, Qwest and WorldCom had losses or fraud-distorted earnings.

then, and in both eras the case for capacity rests on demand that hasn't fully shown up yet: internet traffic then (actual data traffic growth met less than 1% of the capacity laid[4]) and AI efficiencies and enterprise adoption now.

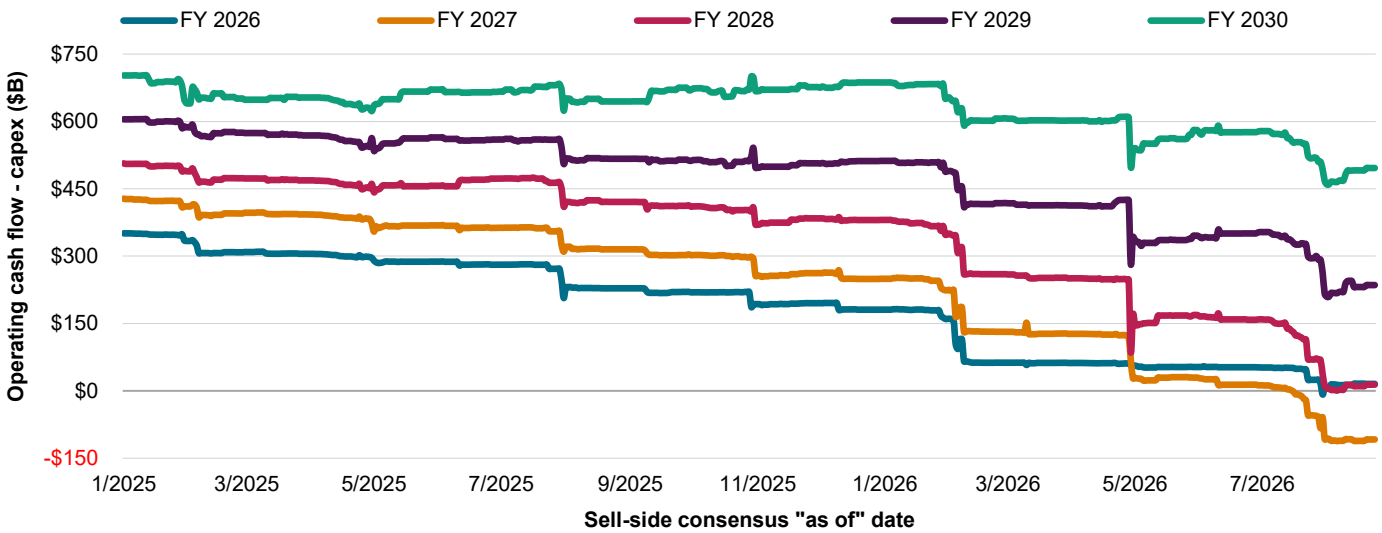
Where the two eras diverge is earnings and how the spending is funded. The dot-com carriers were largely pre-revenue or unprofitable, funding a fiber build-out almost entirely with debt, so when demand disappointed there was no income to fall back on, no cushion to absorb the shortfall, and bankruptcies ensued. The Hyper 5 are at the opposite starting point: five profitable companies funding the build-out almost entirely from their own cash flow; at least so far. Figure 4 shows Visible Alpha Estimates show combined capital spending reaching roughly $735 billion this year and crossing $1 trillion in 2027, an 8x increase from 2021.

That capex is estimated to come from all of the group's operating cash flow this year and to run slightly ahead of it in 2027 (Figure 6), and free cash flow margin, close to 16% of revenue in 2021, compresses to roughly breakeven this year and turns modestly negative in 2027. A strain, but only margin compression on a profitable earnings base rather than the absence of earnings that defined the telecom carriers. Alphabet and Microsoft stay net cash and free-cash-flow positive through 2027. Amazon and Meta swing to modest net debt with free cash flow turning negative. Oracle is the outlier, with net debt above 2.9x earnings and free cash flow estimated at roughly -$46 billion in 2027.

Figure 6

Sell-side consensus now expects capex to outpace cash flow in FY 2027 for S&P 500 Index

Visible Alpha sell-side consensus cash flow minus capex revisions for Hyper 5



As of Aug. 21, 2026.
Source: S&P Global Market Intelligence.
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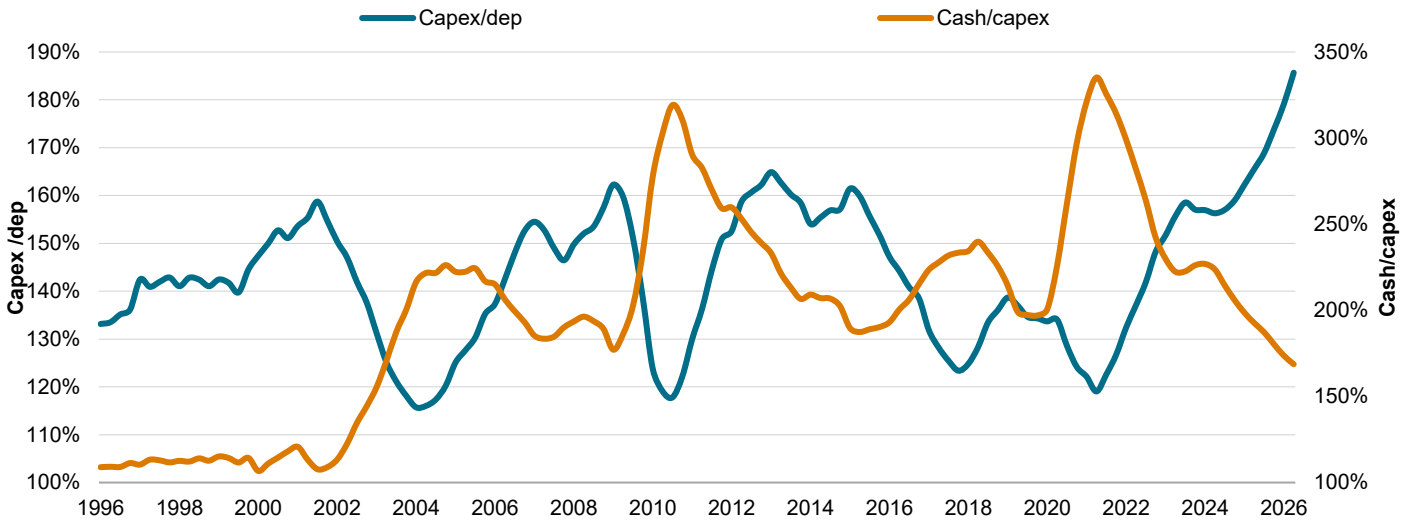
The financing contrast shows up index-wide (Figure 7). S&P 500 cash relative to capex sat around 105% at the 2000 peak, an ordinary position, not a thin one, funded on debt regardless. It climbed to roughly 330% in 2021 on pandemic-era earnings and stimulus, the richest cushion in the 25-year series, and has been drawn down to 150%-160% since, funding a real share of the AI build-out from cash rather than fresh borrowing.

[4] Andrew Odlyzko, University of Minnesota Digital Technology Center — "Measurements and Mismeasurements of the Dynamics of Data Traffic Growth" (2002). users.cse.umn.edu/~odlyzko. [RESEARCH] Between 1998 -2001, U.S. long-haul fiber capacity grew roughly 500-fold whereas actual data traffic demand grew fourfold.

Figure 7

Pandemic-era cash comes off the sidelines as AI capex scales to 50-year high

Bottoms-up capex-to-depreciation and cash-to-capex ratio, S&P 500 (Q1 2000–Q2 2026)



As of Aug. 24, 2026.
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3. What to watch

The key risk in this capex build-out is the transition from cash-funded investment to debt-funded investment before returns on AI infrastructure validate the spending.

Capex exceeding operating cash flow. Group capex is estimated to consume nearly all of Hyper 5 operating cash flow in 2026 and now run slightly ahead of it in 2027. If that ratio holds above 100%, the "cash-funded" distinction from the dot-com era starts to erode mechanically.

The financing mix continuing to shift toward debt. Net new borrowing across the Hyper 5 has already accelerated sharply from a low base. A continued rise in net debt/EBITDA, currently modest in aggregate, would be the clearest indicator the group is financing the build-out with leverage, the way the telecom carriers did.

The cash/capex ratio falling through its 2000 starting level. The index entered this cycle with the richest cash cushion in 25 years and has been drawing it down. Right now, balance sheets look materially different from the dot-com period, but as cash comes off the sidelines that gap is narrowing.

Cost of capital. Right now, the Fed likely has less direct influence over corporate capex than it did in the dot-com era. In a simple capital-cycle frame, the cycle becomes fragile when reported capex begins to persistently exceed operating cash flow, forcing firms to fund the gap with debt and in particular when the system has already drawn down its cash buffer. Once liquidity is tight enough that refinancing costs rise even modestly, the cost of capital can become a “hair trigger:” incremental projects stop penciling, management revises guidance downward, and the elevated reinvestment regime transitions into a slower, capital-digesting phase.

Sell-side expectations. Equity markets in particular respond to sell-side expectations ahead of actuals. Capex guidance revised up faster than earnings or cash flow guidance, and how that impacts the depletion of cash and need for financing, is the primary focus. A pattern already visible over a single quarter of estimate revisions in Figure 7: Capex forecasts for several of the five companies rose considerably faster than operating cash flow forecasts did. If

that pattern repeats each quarter, the incremental capex deployed from 2027 onward starts to become debt funded rather than cash funded and free cash flow turns negative.

Demand materializing on a normal timeline. The dot-com build-out's core problem was not that the fiber was unnecessary — it eventually was used — but that revenue arrived decades later than the capital structures financing it required. The equivalent test here is whether AI-driven revenue and efficiency gains scale within the expectations that current capex is being underwritten against.

4. Methodology and data

S&P Global's data assets encompass a variety of financial data sources used throughout the paper. Compustat fundamentals provide rich history, enabling us to trace the history of corporate investment back to 1964. CIQ fundamentals and estimates offer a perspective on more recent history with associated estimates for upcoming periods. Visible Alpha data provides a highly granular view into sell-side estimates, including changes in consensus over time.

TTM capex and depreciation values were calculated by summing point-in-time capex and depreciation values for the quarter and the prior three quarters. This approach smooths seasonality while being responsive to change. Quarterly cash values also contain seasonality and were smoothed by averaging the current quarter and the prior three quarters.

In Figure 2, hawkish Fed years are years where the US federal funds effective rate[11] increased by more than half a percentage point. The size of the earnings decline bubble is correlated with the size of the year over year earnings change: small for 0 to (-10%), medium for (-10%) to (-20%), and large for anything lower than (-20%). Positive changes are unmarked.

Valuation is measured as trailing price-to-earnings (market capitalization divided by TTM earnings) on quarterly data from 1995 to 2026. For each company, we express the current multiple as a standard-deviation distance ("sigma") from its own historical mean, using an expanding window (minimum 24 quarters). Aggregate segments (the S&P 500 excluding the Hyper 5 and excluding the telecom cohort) are reconstructed from the index total and the named constituents. The Hyper 5 are valued at their most recent market capitalization; the sign of every sigma result is unchanged under five- and 10-year rolling windows.

Compustat Financials

Compustat Financials provides standardized North American and global financial statements and market data for over 80,000 active and inactive publicly traded companies that financial professionals have relied on for over 50 years.

Compustat allows investment professionals, academic researchers, and industry analysts to combine deep history with robust and consistent data standardization into their research and back-testing to produce valuable insights and generate alpha. With historical data for North America as far back as 1950 and point-in-time snapshots beginning in 1987, Compustat provides insight into company financial performance across many different economic cycles not available anywhere else.

Visible Alpha Estimates and S&P Capital IQ Estimates

The S&P Capital IQ and Visible Estimates dataset combines the forecasts, assumptions and logic from full working sell-side models with a comprehensive global contributor network and deep history to provide the most powerful estimate platform in the market. Through S&P Capital IQ Estimates breadth and Visible Alpha Estimates granularity, the data provide both scale and precision, enabling better forecasting, benchmarking, and decision-making.

Visible Alpha's deep consensus data provide a quick understanding of the sell-side view on a company or industry at an unprecedented level of granularity, timeliness and interactivity, leading to a deeper understanding of what is driving a company's future performance through complete product, segment, and industry forecasts. S&P Capital IQ estimates complement this depth with the breadth of an extensive broker contributor network for a broader consensus view, additional company coverage and historical data back to 1996, allowing for consistent and accurate historical analysis across different regions and economic cycles.

SNL Energy and SNL Sector Financials

SNL Energy provides operational and financial data on over 450 power and mid-stream companies, including power plant location and production, operator revenue, new builds and retirements, transmission, and distribution. This dataset illustrates the changes in US power production back to 1994.

SNL Sector Financials cover industry-specific GAAP and IFRS as-reported financials for over 25,000 current and historical companies globally. The dataset covers five key sectors: energy, financial institutions, real estate, metals and mining, and technology, media, and telecommunications.

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Appendix

1964-2023 capex melt-up phases

Post-Covid capex melt-up: plateau at 160% of dep (2021-2023): Build phase reset fast after a sharp but short-lived pause: Capex fell sharply at the start of the pandemic, but the recovery was swift, with spending rising 18% year over year as delayed projects resumed and commodity prices strengthened. There was no real stretch or overshoot phase in this cycle on its own terms; instead, S&P 500 capex rebounded through 2021-2023 as companies rebuilt inventories, reshored supply chains and began investing in cloud and AI infrastructure, settling near 160% of depreciation, not as a resolution but as the on-ramp into the current AI-driven build phase.

QE "soft landing" capex melt-up: exceeds 160% of dep (2010s): Build stretched unusually long: Ultra-low, post-crisis interest rates reduced corporate hurdle rates and kept capex above 160% of depreciation through most of the decade, with the US shale boom as the primary driver, global upstream oil and gas capex peaking near $584 billion[1] in 2011 and US mining sector investment reaching $228.7 billion in 2014, up 13.1%[2] year over year. Cheap financing and higher oil prices sustained aggressive drilling across the Bakken, Eagle Ford and Permian well past the point a tighter money cycle would normally have forced a stretch phase, illustrating how low rates can extend the build phase rather than trigger the usual turn.

Subprime mortgage and GR capex melt-up: exceeds 160% of dep (2006–2007): Build phase ran on cheap, easy credit: US nonfarm business investment rose 14.4% in 2006[3] alone as low financing costs and rising asset prices fueled spending across real estate, financials and industrials, lifting capex above 160% of depreciation by 2007. The stretch phase was compressed almost to nothing, unlike the multiyear dot-com lag: When the subprime crisis triggered a broader credit freeze in 2008, financing didn't just tighten, it disappeared, and overshoot and digestion arrived simultaneously as corporate investment stalled across the economy. The excess here was concentrated in financial engineering and real estate rather than a single technology sector, but the reversal was every bit as abrupt as the telecom unwind once the capital was gone.

Dot-com capex melt-up: nears 160% of dep (around 2000): Build phase was built on anticipated demand: Telecom and early internet infrastructure firms pushed capex toward 160% of depreciation by 2000, overbuilding fiber-optic networks ahead of bandwidth demand[4] that was still mostly forecast. The stretch phase is the textbook case: The market peaked in March 2000, but investment intensity kept accelerating for another 18 months, peaking around September 2001, as WorldCom, Global Crossing, Qwest and Level 3 kept borrowing tens of billions. Overshoot proved brutal: WorldCom admitted to overstating profits by $3.8 billion while carrying roughly $30 billion of debt before filing what was then the largest US bankruptcy at $107 billion in assets[5], and Global Crossing collapsed separately in a roughly $25 billion bankruptcy[6]. Two trillion in global telecom market value[7] between 2000 and 2002 was lost as utilization never caught up to the capacity built for it.

Every cycle to this point, dot-com included, was a technological build-out: capacity built for a new medium (the internet) rather than for physical output. The two cycles that follow predate that shift. Both were industrial in character, capacity built to make, move, and power physical goods, and both unfolded under a fundamentally different accounting backdrop, where high inflation itself did much of the work of pushing the capex ratio higher.

1973/1979 oil shocks and high-inflation capex melt-up: averages >200% for the decade, peaks at 243%: Build accelerated on real economics: the 1973 OPEC embargo and 1979 Iranian Revolution drove major investment in refining, pipelines, domestic exploration and energy-efficiency upgrades as firms adapted to permanently higher energy costs[8]. The stretch phase was mechanically extended rather than resolved by tightening policy: 7%-13%

inflation[9] kept inflating the ratio even after financing conditions turned, since new capex was recorded in current dollars while depreciation still reflected older, cheaper assets. This pushed the ratio above 200% for the decade and to a peak of 243% in 1981. Overshoot finally arrived through Volcker's disinflation campaign and the 1981-82 recession, and digestion followed as the ratio fell to about 140% by 1982. The economy's longer shift toward services set a new, lower baseline that held for four decades[10].

Post-industrial war capex melt-up: peaks at 216% of dep (1964–1968) Build ran through the back half of the postwar industrial expansion: Manufacturing, automotive, steel and chemical firms kept adding capacity built up during the 1950s boom, financing it easily against strong demand and low borrowing costs, and capex rose to 216% of depreciation by the mid-1960s. The stretch phase arrived quietly rather than sharply. Vietnam War spending and early Great Society programs began lifting inflation even as the build-out continued. Because depreciation still reflected assets bought in cheaper postwar dollars, an increasing share of the elevated ratio was a historical-cost artifact rather than fresh investment.[5]

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- June 2026: The Scarcity Premium: Constrained supply boosts earnings expectations on lower volumes, U.S. & LatAm lead: Since Jan. 1, sell-side analysts have lowered observable energy production expectations by 1.9% (1.5 million boe/d1) while raising energy earnings forecasts by 57% ($227 billion). This divergence signals a market revaluation consistent with a scarcity premium: higher profits despite lower expected output. AI adds a related transmission channel. As AI-linked growth increases demand for energy and other constrained inputs, scarcity in one part of the system can manifest as cost pressure elsewhere. At the same time, broader markets have been comparatively stable over the first five months of 2026.
- May 2026: The Most Wanted Exposure: LP Allocation Intent and Unicorn AI Deal Concentration in Private Markets: Capital is flooding into AI, with no end in sight. More than 75% of limited partners (LPs) say they plan to deploy capital into AI over the next 12 months, a stark contrast to blockchain at 18%. LP demand for AI exposure is broad-based: endowments, wealth managers and family offices are all leaning in. But scalable exposure is narrowing. Global AI investment jumped 115% year over year in 2025 to $235 billion from $109 billion, with 82% of the incremental dollars coming from rounds of at least $1 billion. AI is the most-wanted exposure in private markets. It's also becoming the hardest to access without paying up.
- February 2026: Sell-side signals short-term pain, long-term gain: $1T+ in additional costs hit 2025 margins, AI buoys profitability beyond: Global margin expectations were revised lower in the first half of 2025 as sell-side analysts priced in more than $1 trillion of unanticipated costs. By mid-year, the reset was largely complete. Fiscal year 2025 margin growth fell from 80 bps to 45 bps — still positive year over year, but with a decisive 35 bps haircut locked in as fourth-quarter earnings season progressed. The out-years tell a different story. Forward earnings expectations traced a U-shaped recovery, but one that was highly concentrated: just 10 companies account for the majority of the rebound in earnings expectations through 2026 and beyond.
- December 2025: In It Together: How Company Performance Transmits Through Supply Chains: Supply chains transmit information before markets price it. Customer performance reveals emerging demand, and supplier performance reveals capacity and execution risk. These directional spillovers move through networks in predictable ways. When customers are getting stronger, suppliers see order flow and revenue visibility improve. When suppliers are stable and delivering reliably, customers maintain margins and avoid operational bottlenecks. Both channels matter. When both directions turn positive at the same time, the effect compounds: companies benefit from improving demand and a supply base capable of supporting it.
- October 2025: Margin math: A $1.2 trillion expense propagates through supply chains in 2025: Companies are now expected to pay at least $1.2 trillion more in 2025 expenses than was anticipated on Jan. 1. Revenue expectations have risen — but earnings expectations have fallen — producing a 64-basis-point contraction in margin, according to 15,000 sell-side analysts contributing to the S&P Capital IQ and Visible Alpha Estimates. If the pattern holds for firms without sell-side coverage, the cost shock would exceed $1.2 trillion in lost profit.
- September 2025: Familiar signal, new context: Large language models (LLMs) have garnered widespread attention for their ability to understand natural language. Their application in equity investing, however, remains in its infancy due to novelty and cost. This study leverages LLM-extracted features from earnings call transcripts and transforms them into actionable stock selection signals. These features are significantly correlated with their traditional (rules-based) natural language processing counterparts — the lexicon-driven approach that flags and

scores individual phrases as positive or negative — confirming that both approaches measure the same 'ground truth'. The added computational cost and opaqueness of the LLMs appear justified: a fine-tuned LLM-based sentiment strategy would have delivered twice the long-short return performance (LLM 8.4% versus lexicon 4.2%), with a notable advantage in recent years as mispricing opportunities have narrowed.

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